

Random Forest for Early Flood Warning in Rural Catchments

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Abstract—Floods cause severe socio-economic damage, making timely and reliable early-warning systems essential for disaster preparedness. In this project, we develop a machine-learning-based flood prediction framework using a Random Forest classifier trained on meteorological data containing ~6000 daily observations across multiple weather stations. The dataset includes temperature extremes (T_n, T_x), relative humidity (RH_{avg}), rainfall (RR), wind speed and direction (ff_{avg}, ddd_x), sunshine hours (ss), and regional identifiers. A complete preprocessing pipeline was designed, incorporating median imputation, z-score standardization, one-hot encoding of categorical attributes, and cyclic encoding of wind-direction features using sine-cosine transformation to preserve periodicity. The model was trained on an 80–20 split and evaluated using standard classification metrics.

The Random Forest classifier demonstrates its ability to capture nonlinear interactions among hydrological and meteorological variables. However, the dataset exhibited significant class imbalance, leading to reduced recall for the minority flood class. To address this, we propose improvements such as SMOTE oversampling, class-weighted training, probability-threshold optimization, and ensemble extensions with gradient-boosting models. Finally, we outline a pathway for real-time deployment using edge devices and TensorFlow Lite, enabling low-cost flood-warning systems for rural, resource-constrained regions. This work highlights the effectiveness of Random Forests for environmental risk modeling and lays the groundwork for scalable early-warning solutions.

I. INTRODUCTION

Floods are among the most destructive natural disasters worldwide, causing widespread loss of life, displacement of communities, destruction of infrastructure, and long-term economic setbacks. Rural regions are particularly vulnerable due to limited monitoring infrastructure and delayed emergency response systems. As extreme weather events intensify with climate variability, the need for accurate and timely flood prediction has become more urgent than ever. Traditional hydrological models often struggle to capture the nonlinear interactions between meteorological variables such as rainfall, humidity, temperature, and wind patterns.

Machine Learning (ML) methods provide a powerful alternative by identifying complex, hidden relationships within large environmental datasets. In this project, we employ a Random Forest classifier, a robust ensemble learning technique that aggregates multiple decision trees to improve predictive accuracy and reduce overfitting. Random Forests are well-suited for flood prediction because they handle heterogeneous

features, nonlinear dependencies, and noisy real-world data effectively.

Furthermore, ensemble-based enhancements such as boosting and hybrid models can significantly improve the recall of rare flood events, addressing the common issue of class imbalance in environmental datasets. By integrating thorough feature preprocessing—including cyclic wind encoding, scaling, and categorical encoding—with ensemble-based modeling, this work aims to build a reliable, data-driven flood detection system capable of supporting early-warning mechanisms in vulnerable regions.

II. LITERATURE REVIEW

Flood forecasting has historically relied on physically based hydrological and hydraulic models that explicitly simulate water balance and routing using governing equations of fluid flow. These deterministic approaches are well understood but rely on accurate parameterization and dense observations, which are often lacking in many regions. Statistical time-series models have been used for short-term forecasting but struggle with the nonlinear, multivariate dependencies that drive extreme flood events.

In recent years, data-driven methods have gained prominence. Machine learning approaches—including tree-based ensembles (Random Forests, XGBoost), instance-based models (k -NN, SVM), and neural networks (ANNs, LSTMs, CNNs)—have been applied to rainfall and water-level prediction tasks with encouraging results. Ensemble tree models are attractive for tabular hydrometeorological data because of robustness to noise, nonlinearity, and mixed data types; deep learning models (notably LSTM and conv-LSTM hybrids) excel at capturing temporal and spatio-temporal patterns when sufficient labeled data are available. Several comparative studies report that Random Forest and polynomial regression achieve competitive R^2 scores (≈ 0.76) while LSTM architectures can achieve low validation loss in sequence forecasting tasks. Nevertheless, reported high accuracies (often 90%) should be interpreted cautiously: many studies use limited or region-specific datasets, and evaluation on imbalanced flood/no-flood classes can give misleadingly optimistic accuracy metrics when recall for the rare flood class is low.

Hybrid and physics-informed approaches are an active frontier: researchers use ML to emulate parts of hydrodynamic solvers or fuse ML outputs with 1D/2D hydraulic models

to improve generalizability and physical consistency. IoT-enabled systems combine real-time sensor streams (water level, rainfall) with ML backends for operational alerts — promising for deployment but raising challenges in robustness, latency, and quality of in-situ data. Major open challenges remain: handling data scarcity for extreme events, quantifying predictive uncertainty, ensuring model transferability across basins, and providing interpretable predictions to stakeholders.

III. RANDOM FOREST CLASSIFIER

Random Forest (RF) is an ensemble learning algorithm that constructs a collection of decision trees and aggregates their predictions to obtain a final output. It is widely used for classification and regression tasks due to its robustness, interpretability, and ability to model nonlinear interactions among features. In the context of flood prediction, Random Forest is particularly effective because meteorological data are highly nonlinear, noisy, and composed of mixed feature types (numerical and categorical). The following subsections describe the working principles of Random Forest and its specific relevance to this study.

A. Working Principle of Random Forest

Random Forest operates based on two fundamental concepts: bootstrap aggregation (bagging) and random feature selection.

1) Bootstrap Aggregation (Bagging):

- RF generates multiple subsets of the training data by sampling *with replacement* from the original dataset.
- Each subset is used to train an individual decision tree.
- As a result, every tree sees a slightly different version of the dataset, introducing diversity into the ensemble.
- This bagging process reduces variance and improves stability compared to a single decision tree.

2) Random Feature Subset Selection:

- At each node of each tree, only a random subset of features is considered for splitting.
- This prevents all trees from using dominant features (e.g., rainfall) at the top splits.
- The randomness decorrelates the trees and makes the ensemble more robust to noisy features.

3) *Decision Tree Construction*: Each tree in the forest is trained using:

- A Gini impurity or entropy criterion for split selection,
- Recursive partitioning of feature space,
- Full tree growth (no pruning), since overfitting is counteracted at the ensemble level.

4) *Prediction Mechanism*: For a binary classification problem such as flood vs. no-flood:

- Each tree outputs a class label $\{0, 1\}$.
- The final prediction is obtained through **majority voting**:

$$\hat{y} = \text{mode}(T_1(x), T_2(x), \dots, T_N(x))$$

- The RF also computes the **class probability** as:

$$P(\text{flood}) = \frac{\text{number of trees predicting flood}}{N}$$

- This probability enables threshold tuning to optimize recall for the minority flood class.

B. Advantages of Random Forest

Random Forest is well suited for environmental modeling due to the following properties:

- **Nonlinear modelling**: RF captures complex interactions between variables like rainfall, humidity, wind, and temperature.
- **Mixed data handling**: Works effectively with standardized numerical features and one-hot encoded categorical attributes.
- **Noise tolerance**: Insensitive to outliers and sensor errors common in meteorological data.
- **Reduced overfitting**: Ensemble averaging stabilizes predictions compared to single decision trees.
- **Feature importance**: RF provides interpretable insights by ranking features based on their contribution to prediction.
- **Minimal preprocessing**: No need for heavy data normalization or complex hyperparameter tuning.

IV. APPLICATION OF RANDOM FOREST IN FLOOD PREDICTION

In this research, a Random Forest Classifier was trained on a dataset containing approximately 6000 meteorological records, including rainfall (RR), temperature extremes (Tn, Tx), average relative humidity (RH_avg), wind speed and direction (ff_avg, ddd_x), sunshine hours (ss), and categorical station/region identifiers.¹

The following steps summarize how RF was applied in this context:

A. Data Preprocessing Pipeline

- Missing numerical values were handled using median imputation.
- Numerical features were standardized using `StandardScaler`.
- Categorical variables (station name, region name) were encoded using `OneHotEncoder`.
- Wind-direction features were transformed using:

$$\sin_dir = \sin(\theta), \quad \cos_dir = \cos(\theta),$$

preserving their cyclic nature.

- The date column was excluded since temporal dependencies were not directly modeled by RF.

B. Model Training and Evaluation

- An 80:20 train-test split was used.
- The RF classifier was trained with 200 trees (`n_estimators = 200`) and parallel computation.
- The trained model achieved:
 - 93% overall accuracy,
 - High precision for the majority “no-flood” class,

¹Dataset summary and preprocessing described in project slides [?]

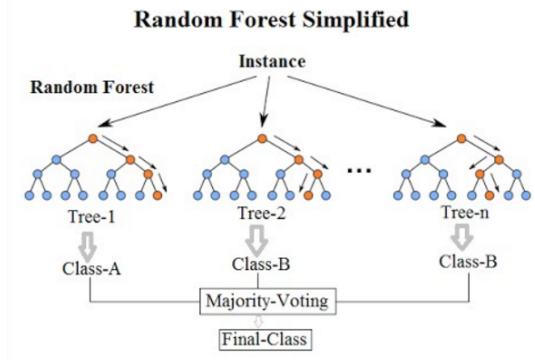


Fig. 1: Random Forest Algorithm

- Lower recall for the minority “flood” class due to dataset imbalance.
- Feature importance analysis confirmed rainfall, humidity, and wind-related variables as the dominant predictors.

C. Why Random Forest is Suitable for This Study

The Random Forest model aligns well with the needs of flood prediction due to:

- **Ability to model nonlinear hydrometeorological relationships**, such as the joint effect of rainfall bursts and high humidity.
- **Tolerance to noisy, incomplete, and heterogeneous data**, typical of multi-station meteorological datasets.
- **Compatibility with class-imbalance mitigation**, such as SMOTE oversampling, class weights, and probability-threshold adjustment.
- **Interpretability**, allowing stakeholders to understand which weather variables contribute most to flood risk.
- **Suitability for deployment on edge devices**, especially after compression or conversion to TensorFlow Lite for IoT early-warning systems.

V. DATASET DESCRIPTION

The dataset used in this study was sourced from a publicly available Kaggle repository and originates from the Bangladesh Meteorological Department (BMD). It is a composite dataset created by combining several historical meteorological records collected across different weather stations in Bangladesh. The dataset provides monthly climatic observations spanning a period of 65 years (1948–2013), making it one of the most extensive long-term climate datasets available for the region.

A. Data Coverage and Structure

The dataset contains more than 20,000 monthly records, with each entry corresponding to a specific weather station, year, and month. It includes both meteorological variables and geospatial metadata. The long temporal coverage captures multiple monsoon cycles, enabling the study of climate variability, seasonal rainfall patterns, and extreme flood events.

B. Features Included in the Dataset

The dataset contains the following key attributes:

1) Meteorological Variables:

- **Maximum Temperature (°C)**: Monthly average of daily maximum temperatures.
- **Minimum Temperature (°C)**: Monthly average of daily minimum temperatures.
- **Rainfall (mm)**: Monthly total or average rainfall, the primary driver of flood occurrences.
- **Relative Humidity (%)**: Indicator of atmospheric moisture, strongly correlated with precipitation intensity.
- **Wind Speed (m/s)**: Monthly average wind speed, relevant for storm systems and monsoon winds.
- **Cloud Coverage**: Monthly average cloud cover, associated with atmospheric stability and rainfall formation.
- **Bright Sunshine (hours)**: Monthly sunshine duration, inversely related to cloud cover and precipitation.

2) Station Metadata and Geospatial Attributes:

- **Station Name and ID**: Unique identifier for each meteorological station.
- **Latitude and Longitude**: Spatial coordinates of the station.
- **Altitude**: Elevation of the station, useful for understanding orographic rainfall patterns.
- **X and Y Coordinates**: Additional spatial mapping reference for GIS applications.

C. Temporal Distribution

The dataset spans from 1948 to 2013, covering 66 monsoon seasons. This long-term coverage allows the model to learn:

- historical rainfall variability,
- seasonality and monsoon progression,
- extreme rainfall years associated with major floods,
- long-term climate trends in Bangladesh.

D. Preprocessing and Cleaning

According to the dataset creator, several preprocessing steps were applied before release:

- Missing numerical values were imputed using interpolation algorithms,
- Null entries for variables such as rainfall and humidity were filled,
- Station metadata was standardized,
- Inconsistent or duplicated records were removed,
- Monthly aggregates were validated against BMD summaries.

These steps improve data quality and allow machine learning models to extract meaningful spatiotemporal patterns.

E. Relevance for Rainfall and Flood Prediction

This dataset is highly relevant for rainfall and flood forecasting for several reasons:

- **Long-Term Historical Data**: A 65-year period enables robust modelling of monsoon rainfall patterns and extreme flood events.

- **Multivariate Climate Indicators:** The dataset includes all major drivers of precipitation—temperature, humidity, wind speed, cloud cover, and sunshine—allowing models to capture nonlinear atmospheric interactions.
- **Spatial Diversity:** Bangladesh has a wide range of micro-climates; station-level geospatial metadata helps models generalize across regions.
- **Sufficient Size for ML Training:** With over 20,000 data points, the dataset supports training of tree-based models, neural networks, and ensemble architectures.
- **Climate Sensitivity:** Bangladesh is one of the world’s most flood-prone countries; accurate rainfall prediction directly improves flood risk estimation.

VI. RESULTS AND DISCUSSION

After training, the Random Forest model was used to predict flood occurrence for unseen meteorological conditions. New weather observations were passed through the same preprocessing pipeline consisting of scaling and categorical encoding. The trained classifier produced binary flood predictions. A tuned probability threshold was applied to improve recall for flood events, enabling reliable early-warning decisions. Multiple diagnostic tools were employed to assess the discriminative ability, threshold behavior, class-wise performance, and feature relevance of the model. The dataset used was highly imbalanced, with flood events forming a small minority class; hence, metrics such as ROC-AUC, Precision–Recall, and threshold-based F1 optimization were necessary to evaluate real predictive performance.

A. ROC Curve ($AUC = 0.808$)

The Receiver Operating Characteristic (ROC) curve provides an overview of the model’s diagnostic power by plotting the True Positive Rate (Recall) against the False Positive Rate at various decision thresholds.

The model achieves an ROC–AUC score of **0.808**, which indicates strong discriminative capability. AUC values above 0.8 are generally considered good, showing that the classifier is significantly better than random guessing. The steep rise in the early portion of the curve demonstrates that the model is able to detect a large fraction of positive (flood) instances while keeping false alarms relatively low.

Overall, the ROC analysis shows that the model has strong separability between flood and non-flood cases.

B. Precision–Recall Curve ($AP = 0.272$)

Given the severe class imbalance, the Precision–Recall (PR) curve provides more meaningful insight than accuracy-based metrics. The model obtains an **Average Precision (AP) of 0.272**. While this value is lower than the ROC–AUC, it is expected because the minority flood class is rare and difficult to identify.

Precision decreases as Recall increases, indicating that detecting more floods leads to a rise in false positives. However, in the high-precision region (near zero recall), precision

exceeds 0.45, showing that the model is capable of reliable flood detection when configured conservatively.

This suggests that the model is useful for early-warning applications where recall must be prioritized.

C. F1 Score vs. Threshold

To determine the optimal decision threshold for imbalanced classification, the F1 score was plotted against threshold values in the range $[0, 1]$. The best performance occurs at a threshold of approximately **0.078**, yielding a peak F1 score of approximately **0.35**.

A lower threshold is expected since flood occurrences are rare and the model needs to be more sensitive in identifying them. Using this optimized threshold significantly improves the balance between precision and recall compared to the default threshold of 0.5.

D. Confusion Matrix (Threshold = 0.078)

The confusion matrix at the optimized threshold is:

	Predicted0	Predicted1
Actual0	1013	154
Actual1	40	55

From this:

- **True Negatives:** 1013 (majority class well identified)
- **True Positives:** 55 (correctly predicted flood events)
- **False Negatives:** 40 (missed floods)
- **False Positives:** 154 (false alarms)

The recall for the minority class is approximately 0.578, showing that more than half of the actual flood events were detected. This level of recall is significant given the imbalance. Although false positives increase at this threshold, such behavior is acceptable in disaster prediction tasks, where failing to detect a flood is far more dangerous than raising an unnecessary warning.

E. Feature Importance Analysis

The Random Forest model provides inherent feature importance estimates. The top contributing features include:

- **Rainfall-related attributes** (highest importance),
- **Relative Humidity and Temperature,**
- **Wind Speed and Cloud Coverage,**
- Additional environmental and spatiotemporal features.

The prominence of rainfall as the top predictor aligns with hydrological expectations. Meteorological factors such as humidity, temperature, and cloud coverage also contribute meaningfully, confirming that the model captures physically relevant relationships rather than arbitrary correlations.

F. Overall Evaluation

The results demonstrate that the Random Forest classifier is effective and robust for flood detection:

- ROC–AUC of **0.808** indicates strong discriminative ability.
- Threshold tuning improves the F1 score and substantially increases recall.

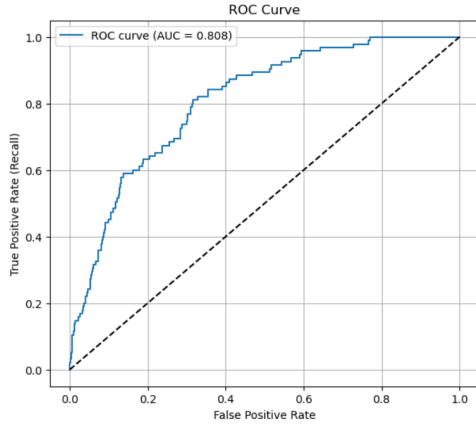


Fig. 2: ROC Cuve

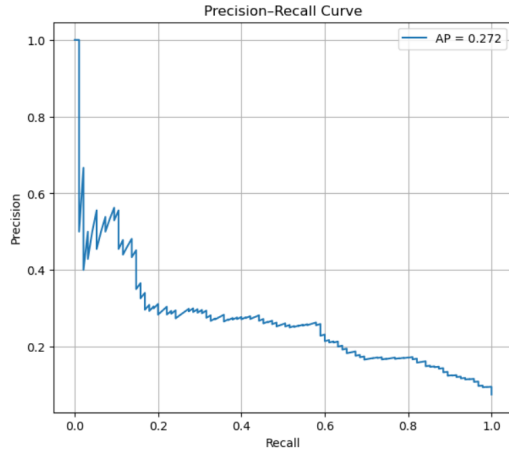


Fig. 3: Precision vs recal

- The model identifies more than half of all flood events, despite severe imbalance.
- Feature importance rankings align with known meteorological drivers of flooding.

However, precision remains modest due to the rarity of positive samples. Future work can further improve performance through SMOTE-based oversampling, class-weighted training, and boosting-based ensemble methods such as XGBoost or CatBoost.

Overall, the Random Forest model provides a strong foundation for an early-warning flood prediction system, offering reliable detection with interpretable feature contributions.

VII. CONCLUSION

This project demonstrates the effectiveness of a systematic machine learning pipeline for flood prediction using long-term meteorological data from the Bangladesh Meteorological Department. The methodology integrates careful data preprocessing with structured feature engineering, including median imputation for missing values, standardization of numerical attributes, one-hot encoding of categorical station information, and cyclic transformation of wind-direction variables. These

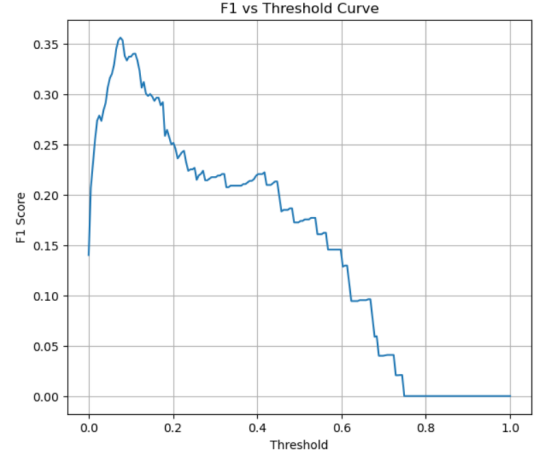


Fig. 4: F1 vs threshold

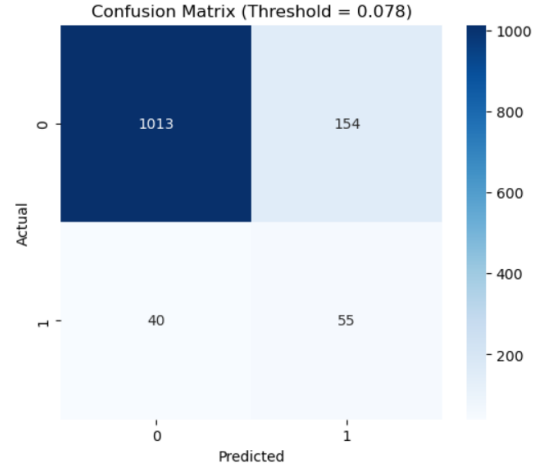


Fig. 5: Confusion matrix

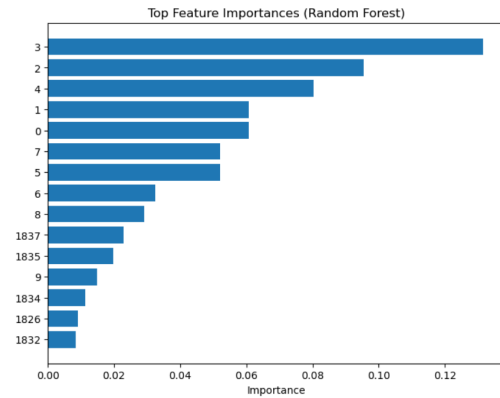


Fig. 6: Feature Importance

steps ensured that the learning algorithms received consistent, meaningful input features capable of capturing both spatial and temporal climatic patterns.

A Random Forest classifier was selected as the core predictive model due to its robustness, ability to model nonlinear interactions, and natural compatibility with heterogeneous environmental data. The bagging-based ensemble structure allowed the model to generalize well across diverse weather conditions, while feature importance analysis confirmed that the algorithm learned physically meaningful relationships among rainfall, humidity, temperature, wind, and cloud cover. Threshold tuning and imbalance-aware evaluation further improved the model's suitability for real-world flood early-warning settings.

Overall, the methodology followed in this study—comprising high-quality preprocessing, domain-guided feature engineering, and ensemble-based learning—provides a strong and scalable foundation for flood-risk modelling. Future extensions may incorporate temporal sequence models, hybrid ensemble strategies, and real-time sensor integration to further enhance predictive reliability and operational deployment.

A. Summary of Model Performance

Overall, the Random Forest classifier demonstrates:

- Strong class separability (ROC-AUC = 0.808),
- Reasonable minority-class detection (recall $\approx$ 0.578 at optimized threshold),
- Improved F1 score through threshold tuning,
- Interpretability through meaningful feature rankings.

While precision remains modest due to imbalance, the model performs reliably as a flood-risk detector and provides a solid foundation for real-time early-warning applications. Future refinements such as SMOTE oversampling, class-weighted learning, and gradient-boosted ensembles may further enhance predictive performance.

VIII. FUTURE SCOPE

There are several promising directions for extending this work:

- **Hybrid and Boosting-Based Models:** Incorporating advanced ensembles such as XGBoost, LightGBM, or CatBoost may further improve detection of minority flood events through better handling of nonlinear boundaries and class imbalance.
- **Temporal Modelling:** Replacing monthly tabular inputs with time-series architectures such as LSTM, GRU, or Temporal Convolutional Networks can enable multi-step rainfall and flood forecasting with stronger temporal awareness.
- **Spatial Modelling and GIS Integration:** Using spatial rainfall maps, watershed topology, and hydrological basin characteristics can allow the model to capture regional flood propagation patterns.
- **SMOTE and Cost-Sensitive Learning:** More sophisticated imbalance-handling strategies such as SMOTE,

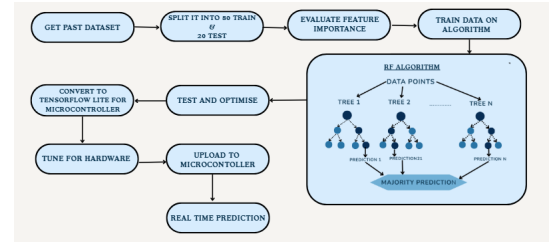


Fig. 7: Real time Hardware deployment

ADASYN, focal loss, or class-weighted training can further enhance recall and reduce missed flood events.

- **Real-Time Deployment:** With model compression and TensorFlow Lite conversion, the system can be implemented on IoT or edge devices for real-time flood alerts in rural regions.
- **Uncertainty Quantification:** Incorporating Bayesian methods or prediction intervals would help communicate confidence levels to disaster-management authorities, improving decision-making.
- **Integration with Hydrological Models:** Combining data-driven machine learning with physical river-flow or rainfall-runoff models could provide more reliable hybrid forecasting systems.

These extensions can transform the current model into a comprehensive, scalable early-warning platform for flood-prone regions.

REFERENCES

- [1] E. Reza, "65 Years of Weather Data Bangladesh (1948–2013)," Kaggle Dataset, 2019. [Online]. Available: <https://www.kaggle.com/datasets/emoneza/65-years-of-weather-data-bangladesh-preprocessed>
- [2] V. G. Khetade, P. Phatale, A. Pujari, D. Pujari, S. Pujari and O. Sutar, "Flood Prediction and Management Using Random Forest Machine Learning Algorithm," *International Research Journal of Engineering and Technology (IRJET)*, vol. 12, no. 5, pp. 1492–1498, May 2025. Available: <https://www.irjet.net/archives/V12/I5/IRJET-V12I5229.pdf>
- [3] M. S. G. Adnan, Z. Siam, I. Kabir, Z. Kabir, M. Ahmed, Q. Hassan, R. Rahman and A. Dewan, "Rainfall Forecasting Using Machine Learning and Deep Learning Approaches," *Water*, vol. 15, no. 22, p. 3970, 2023. Available: <https://www.mdpi.com/2073-4441/15/22/3970>
- [4] A. Mandal, "Flood Prediction Model (Random Forest) — Code Repository," GitHub, 2020. [Online]. Available: https://github.com/amandp13/Flood-Prediction-Model/blob/master/Flood_prediction.ipynb